Prática de Física dos Dispositivos Eletrônicos

FGA0100

Laboratório-6

Efeito Piezoelétrico

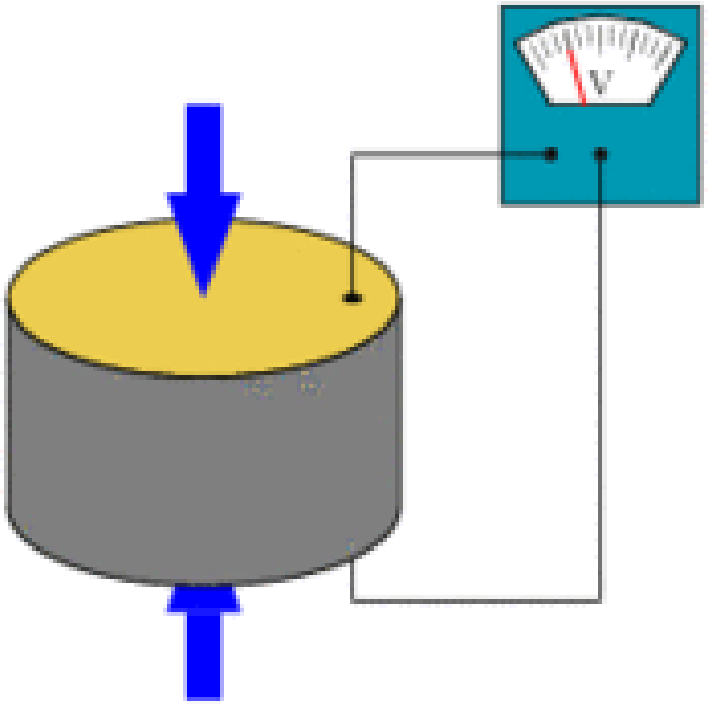
01/2022

FGA

Universidade de Brasília

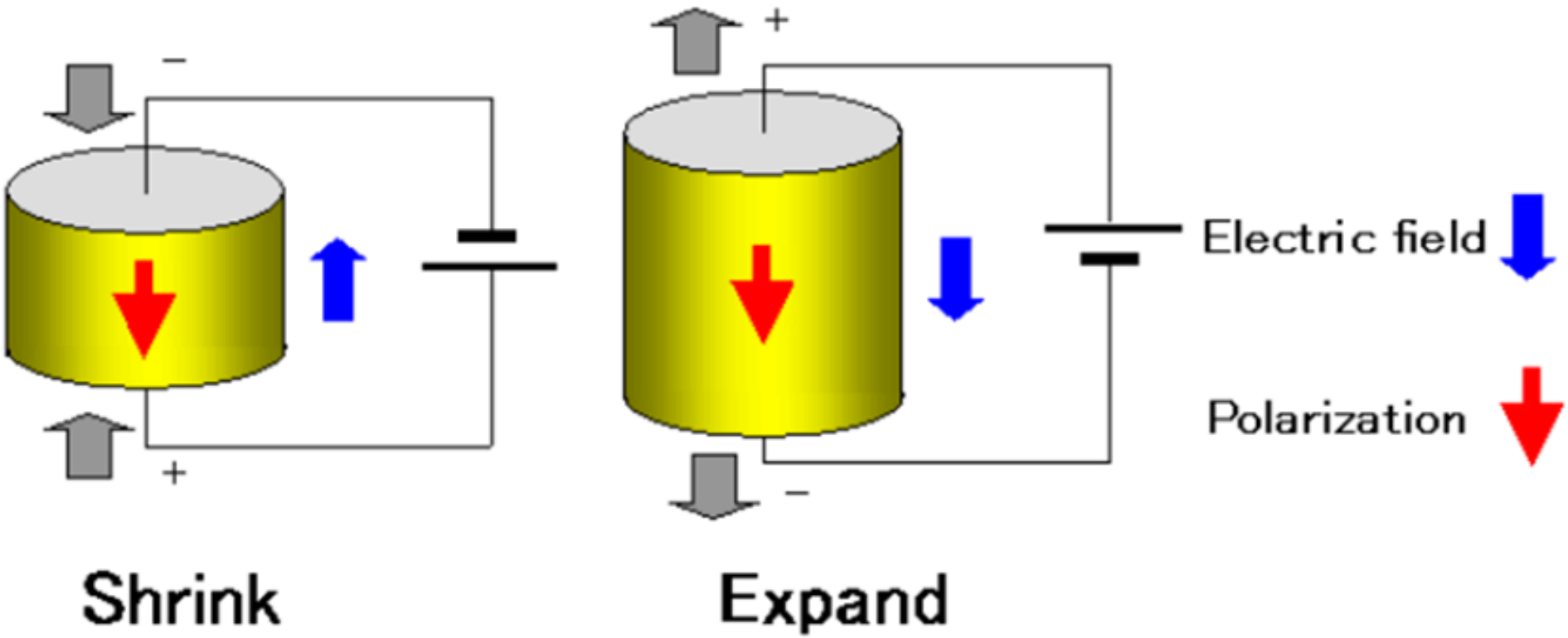
Efeito Piezoelétrico (1880)

Jacques e Pierre Curie



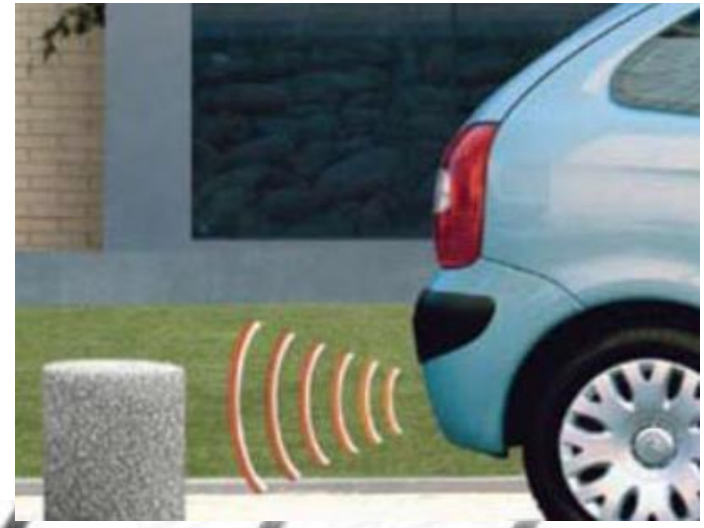
Curie, Jacques and Curie, Pierre (1880). “Development, via compression, of electric polarization in hemihedral crystals with inclined faces,” *Bulletin de la Societe de Minerologique de France*, 3: 90-93.

Efeito Piezoelétrico Inverso (1881)

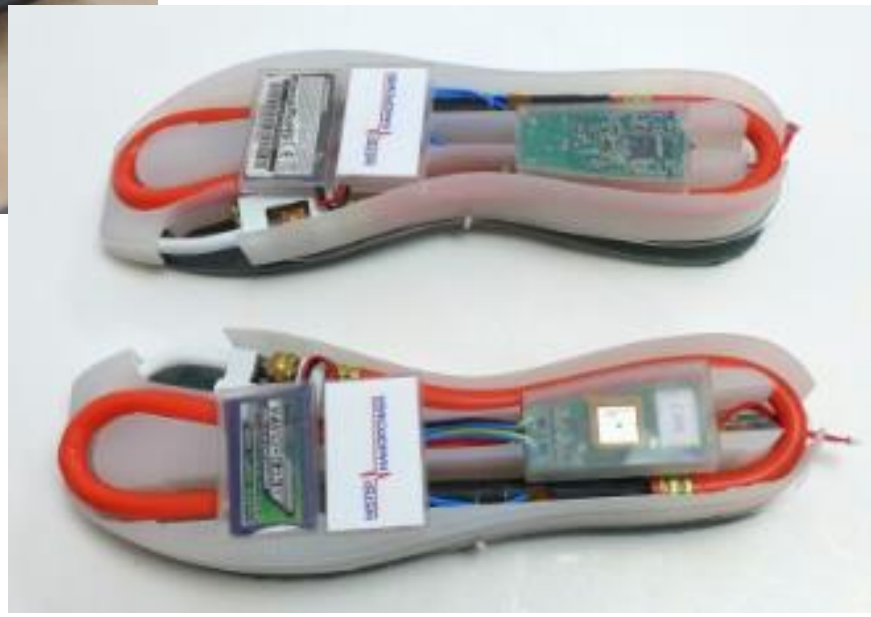


Curie, Jacques, and Curie, Pierre (1881). “Contractions and expansions produced by voltages in hemihedral crystals with inclined faces,” *Comptes Rendus* 93: 1137-1140.

Aplicações de Materiais Piezoelétricos

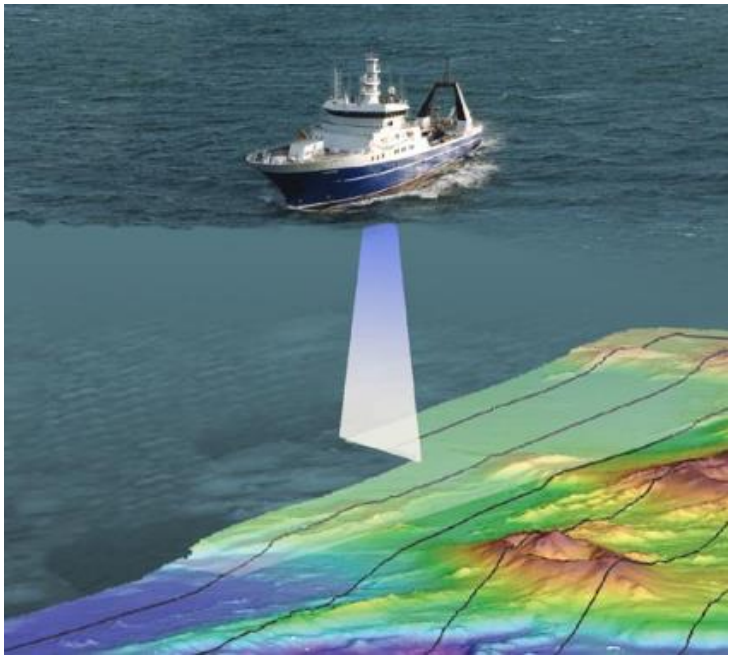


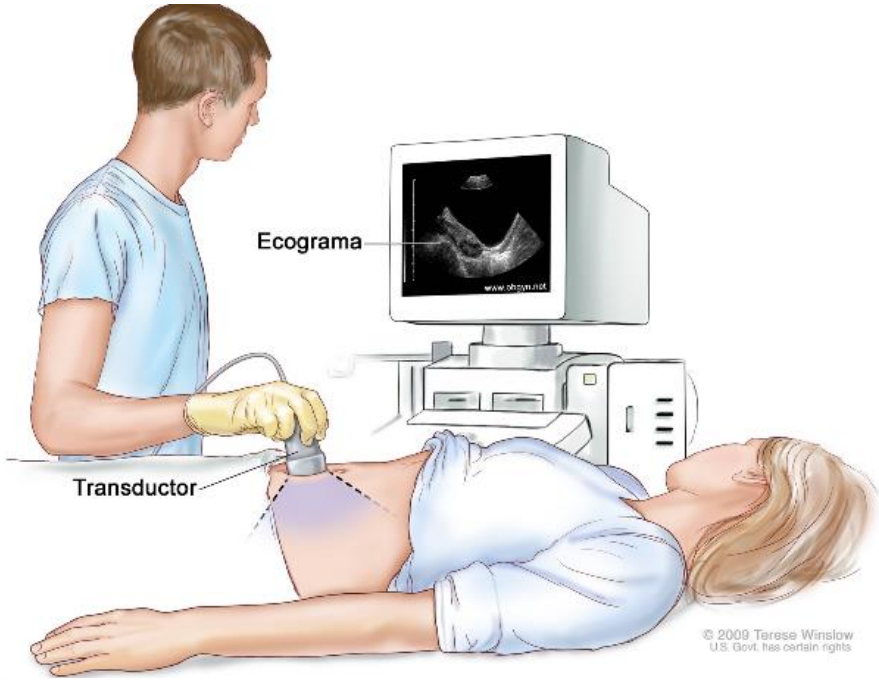
Tênis Piezoelétrico



http://news.wisc.edu/power-walk-footsteps-could-charge-mobile-electronics/

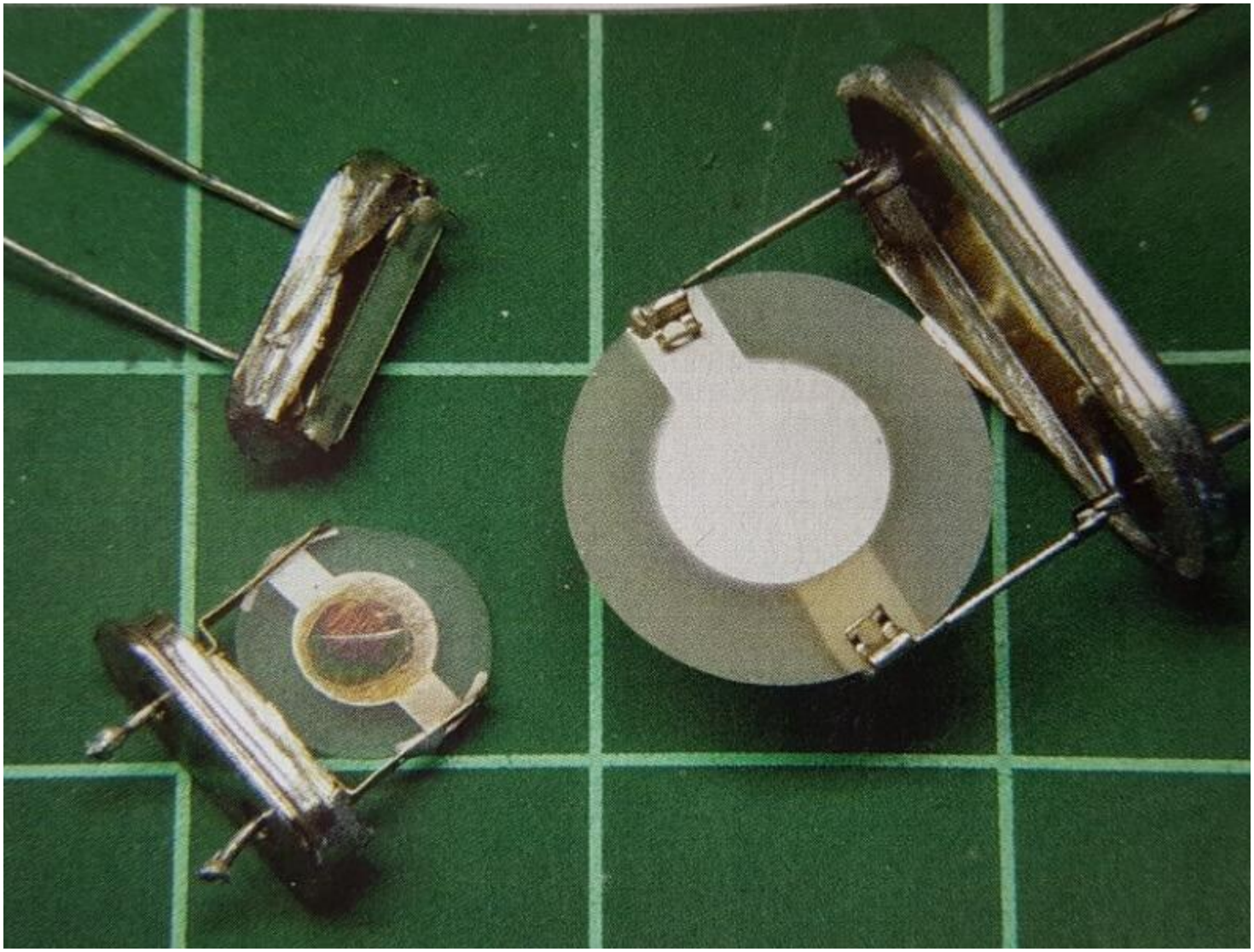
Imagens: Sonar e Ecograma



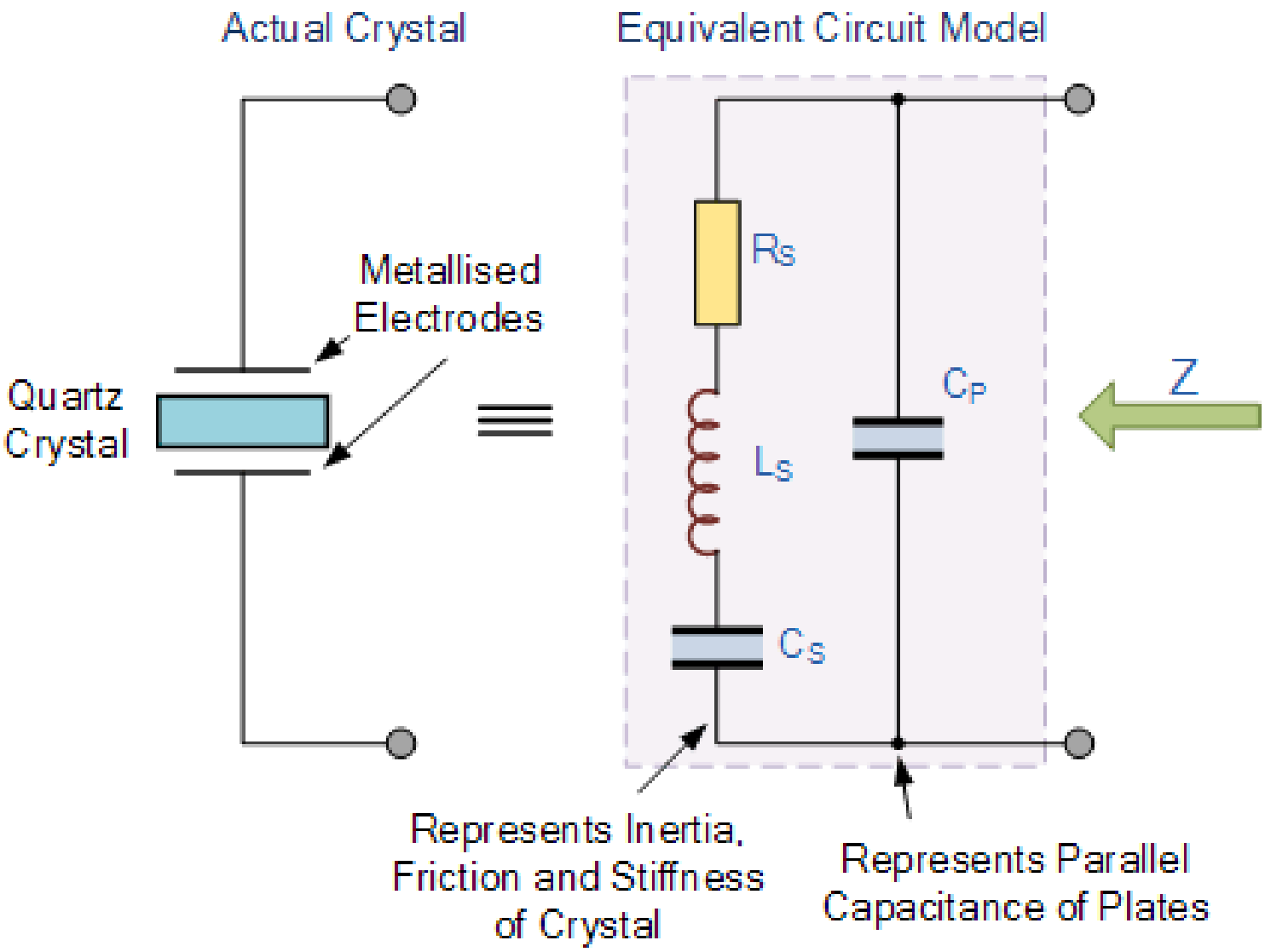


Aplicações Tecnológicas de Materiais Piezoelétricos

Cristal de Quartzo Piezoelétrico

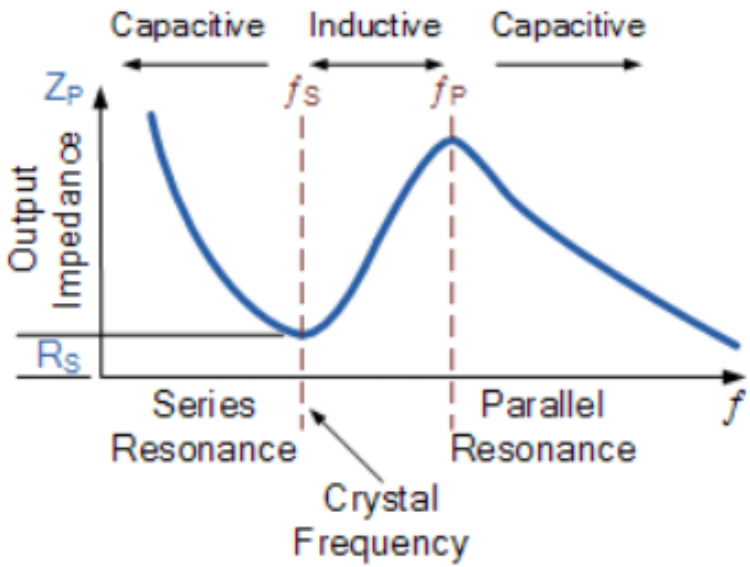


Modelo do Cristal de Quartzo Piezoelétrico



https://www.electronics-tutorials.ws/oscillator/crystal.html

Crystal Impedance against Frequency



$$R = R \quad \text{and} \quad X_{LS} = 2\pi f L_S$$

$$X_{CS} = \frac{1}{2\pi f C_S} \quad \text{and} \quad X_{CP} = \frac{1}{2\pi f C_P}$$

$$Z_S = \sqrt{R_S^2 + \left(X_{LS} - X_{CS}\right)^2}$$

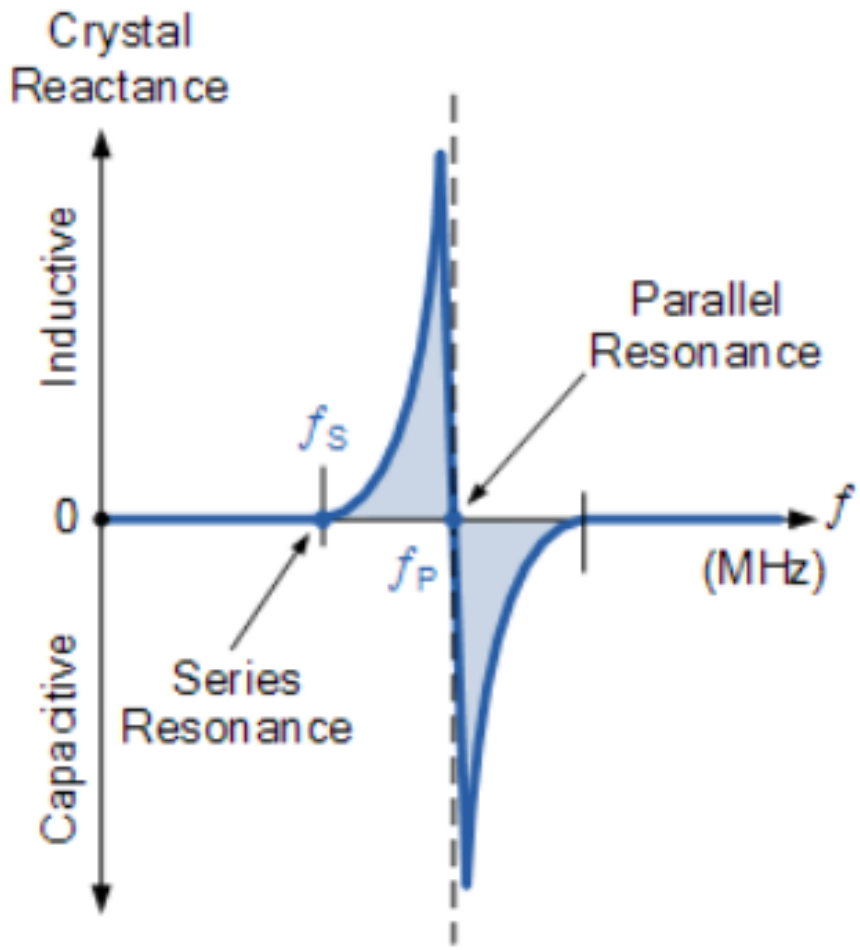
$$\therefore Z_P = \frac{Z_S \times X_{CP}}{Z_S + X_{CP}}$$

$$f_S = \frac{1}{2\pi\sqrt{L_S C_S}}$$

$$f_P = \frac{1}{2\pi\sqrt{L_S\left(\frac{C_P C_S}{C_P + C_S}\right)}}$$

Efeito da Temperatura: $f = f_0[1 - 0.04\ \text{ppm}/^\circ\text{C}^2 \cdot (T - T_0)^2]$

Crystal Reactance against Frequency

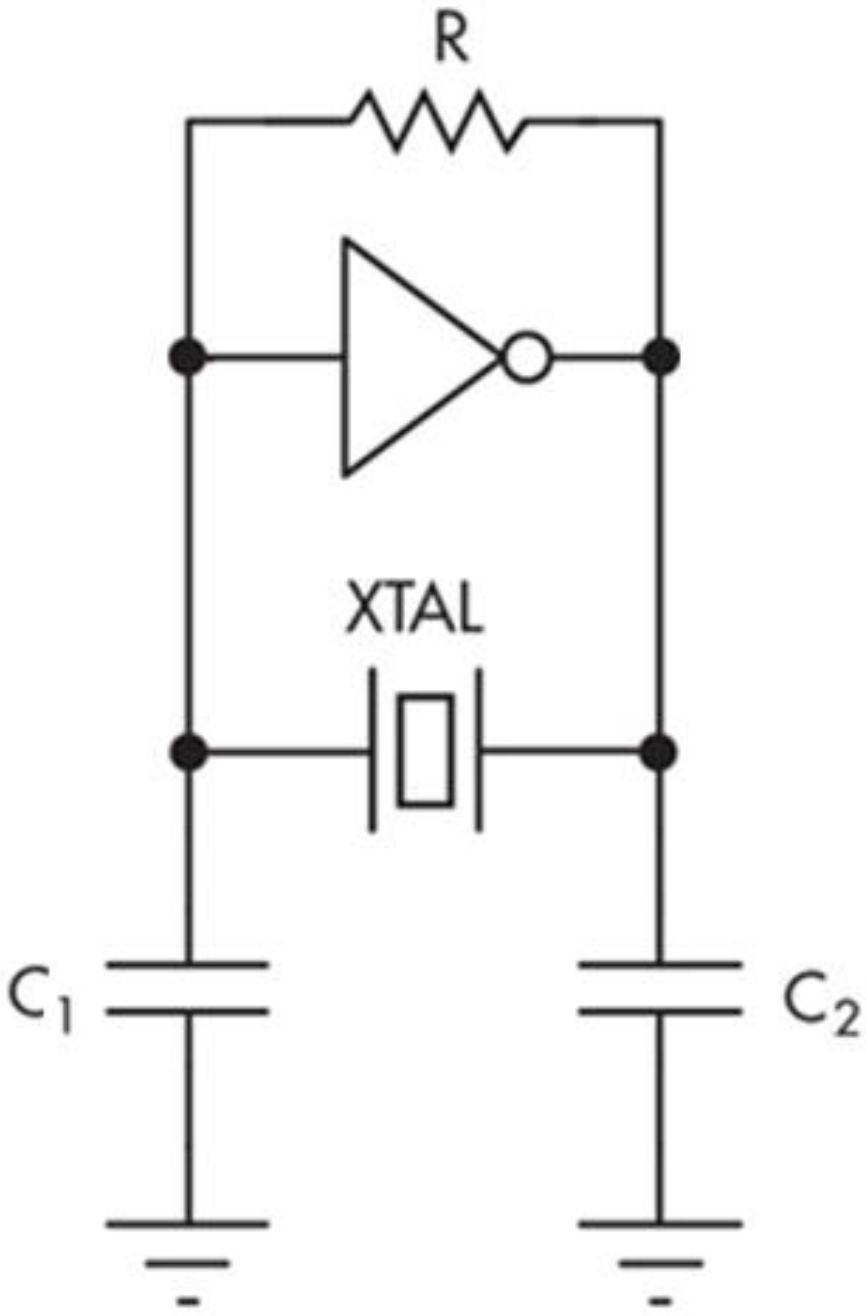


$$X_S = R^2 + \left(X_{LS} - X_{CS}\right)^2$$

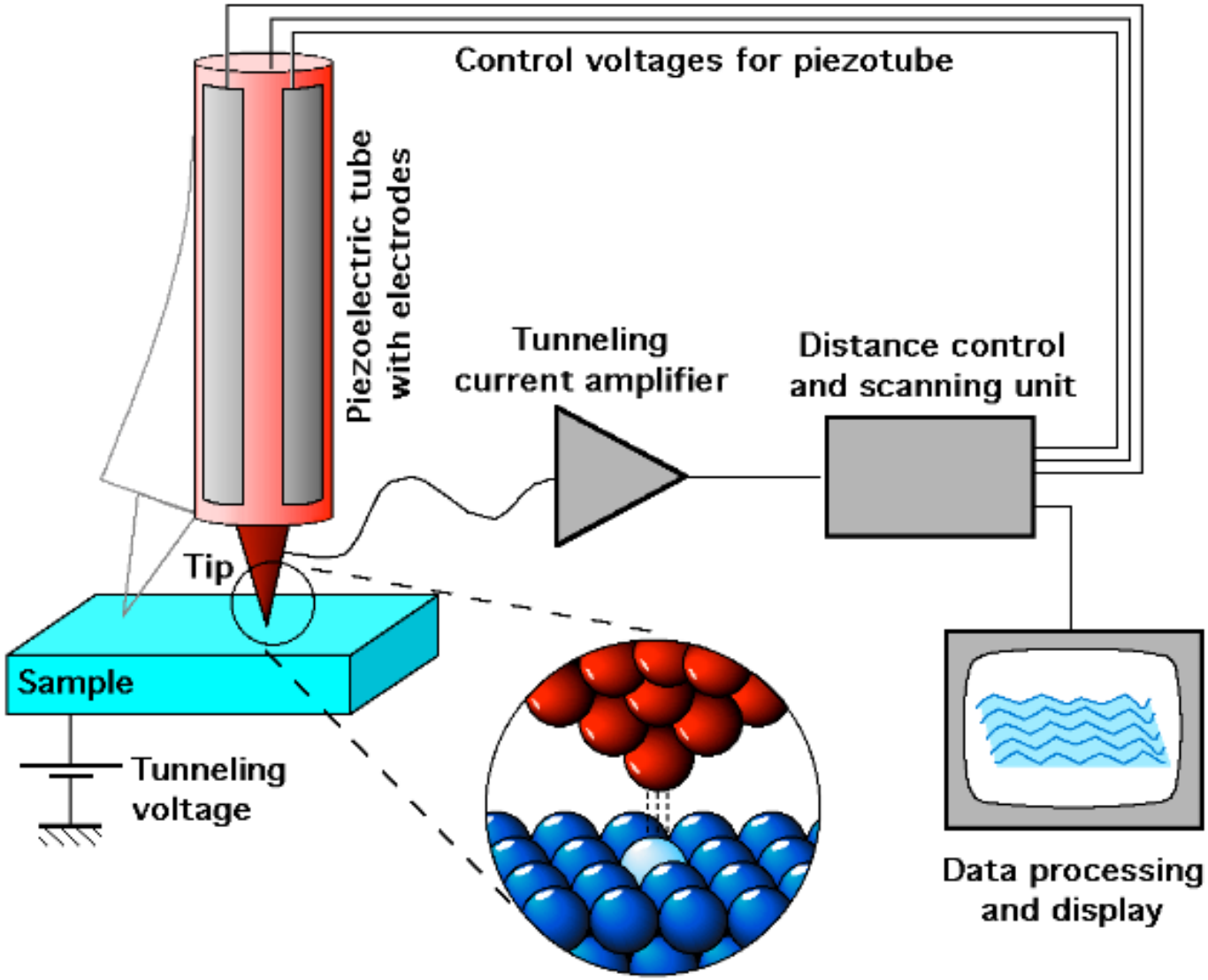
$$X_{CP} = \frac{-1}{2\pi f C_P}$$

$$X_P = \frac{X_S \times X_{CP}}{X_S + X_{CP}}$$

Oscilador CMOS com Cristal de Quartzo Piezoelétrico



STM – Scanning Tunneling Microscope



PZT - Titanato Zirconato de Chumbo

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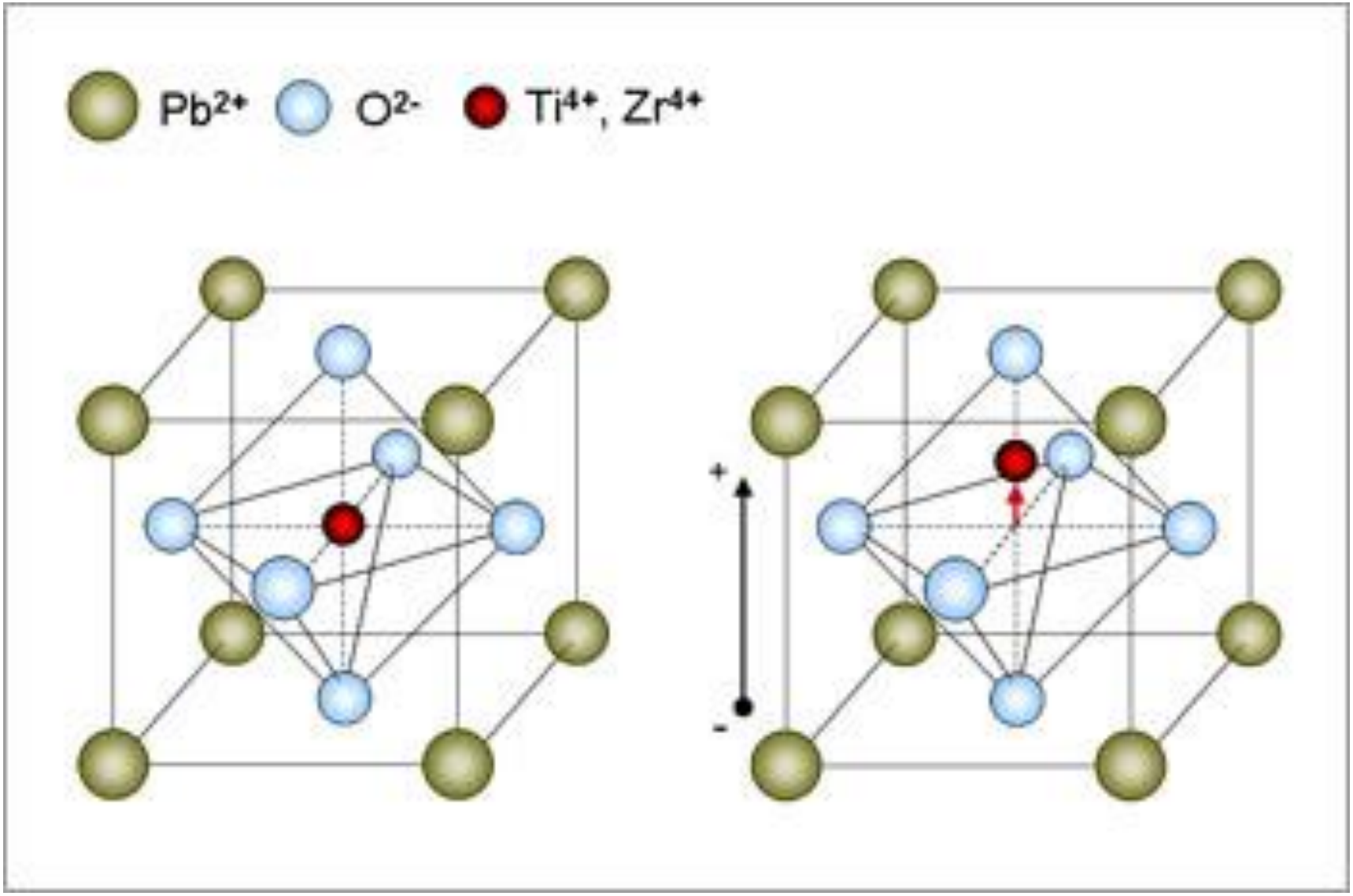
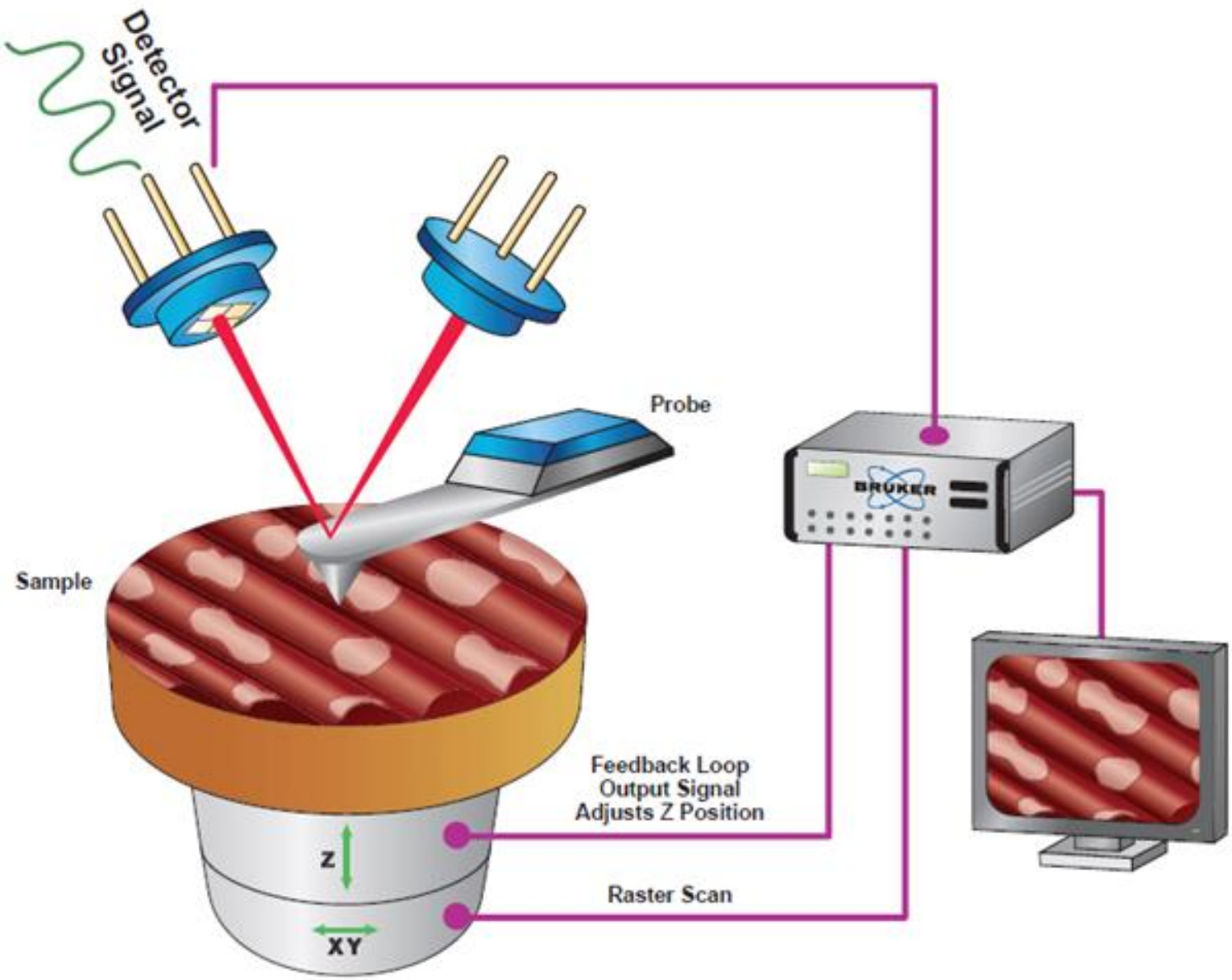
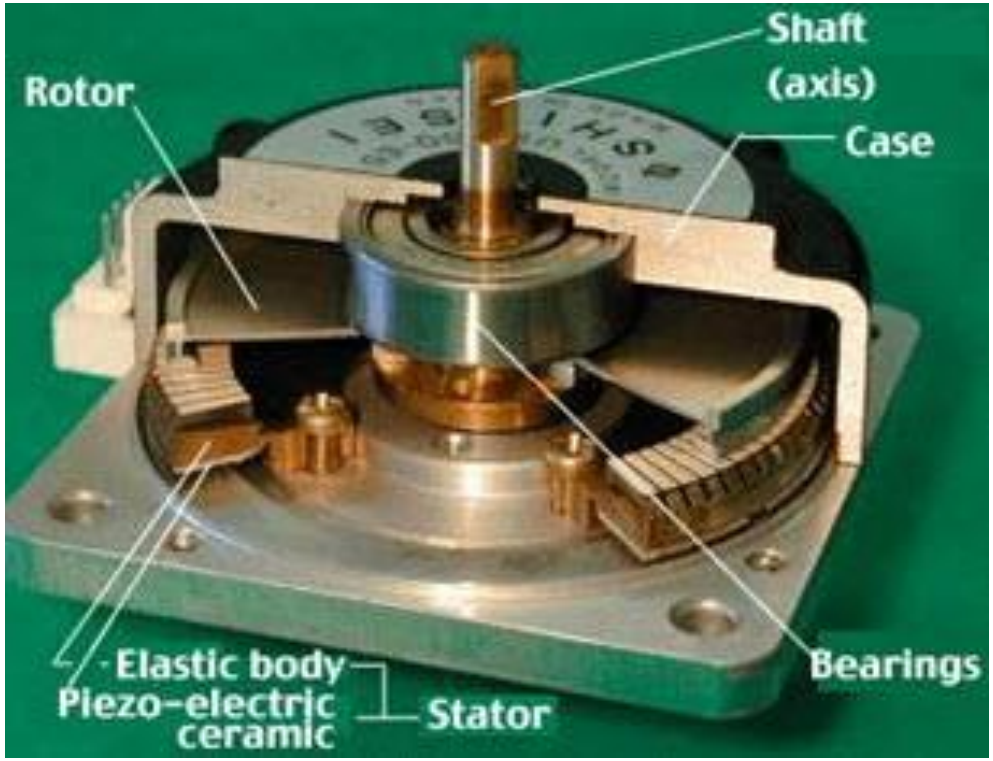
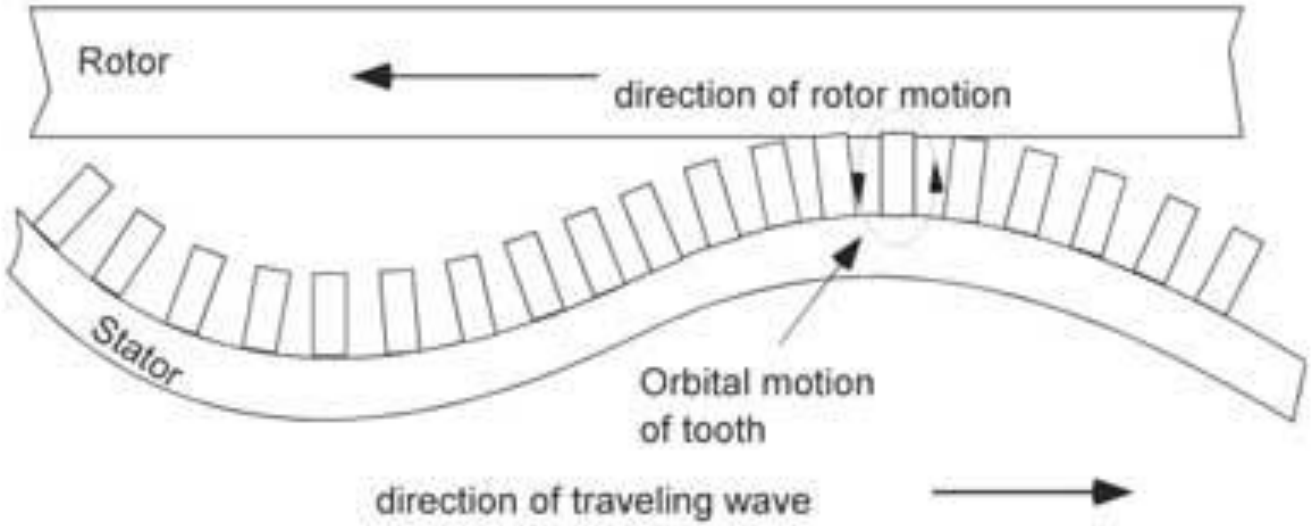


Atomic Force Microscope

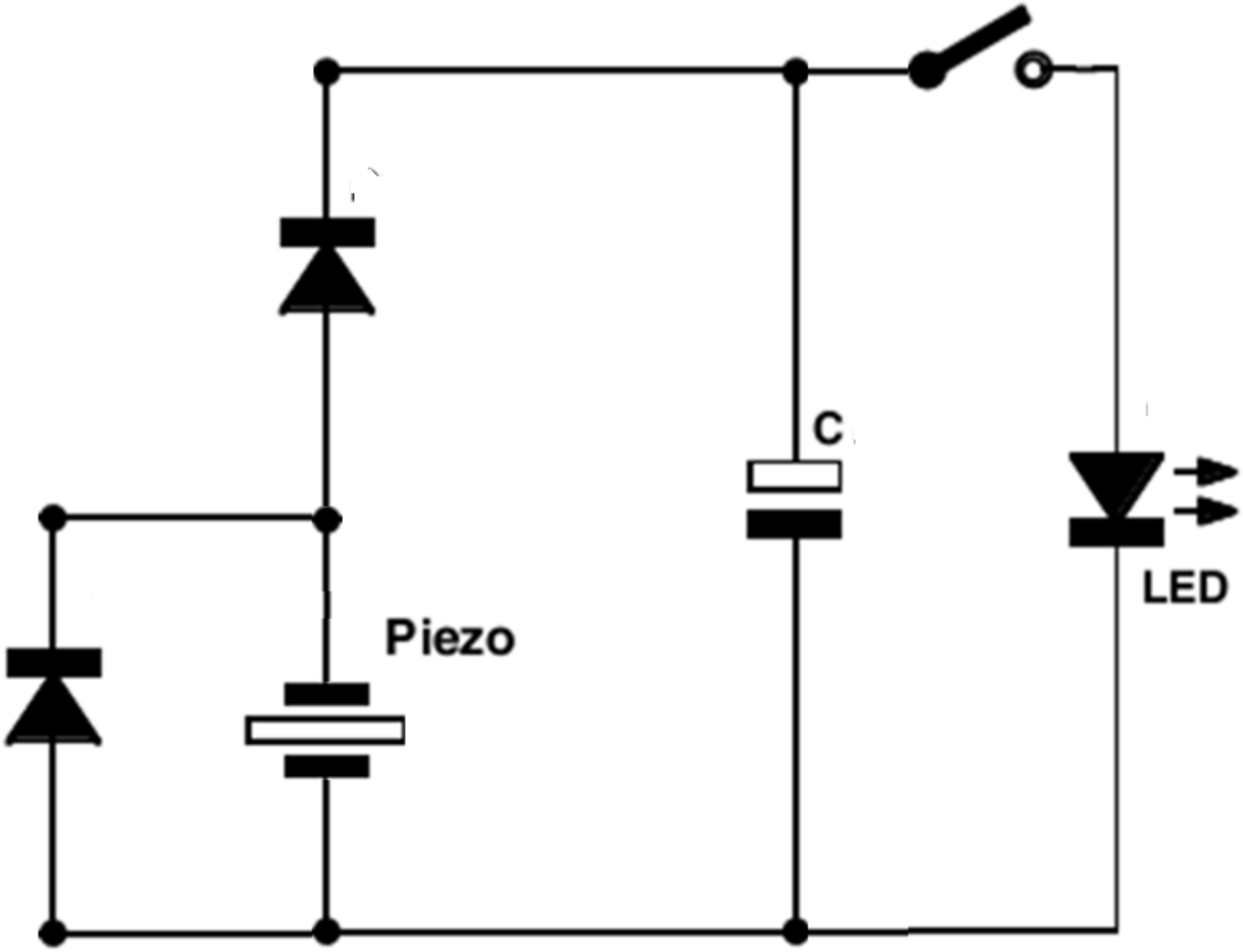


Motor Piezoelétrico

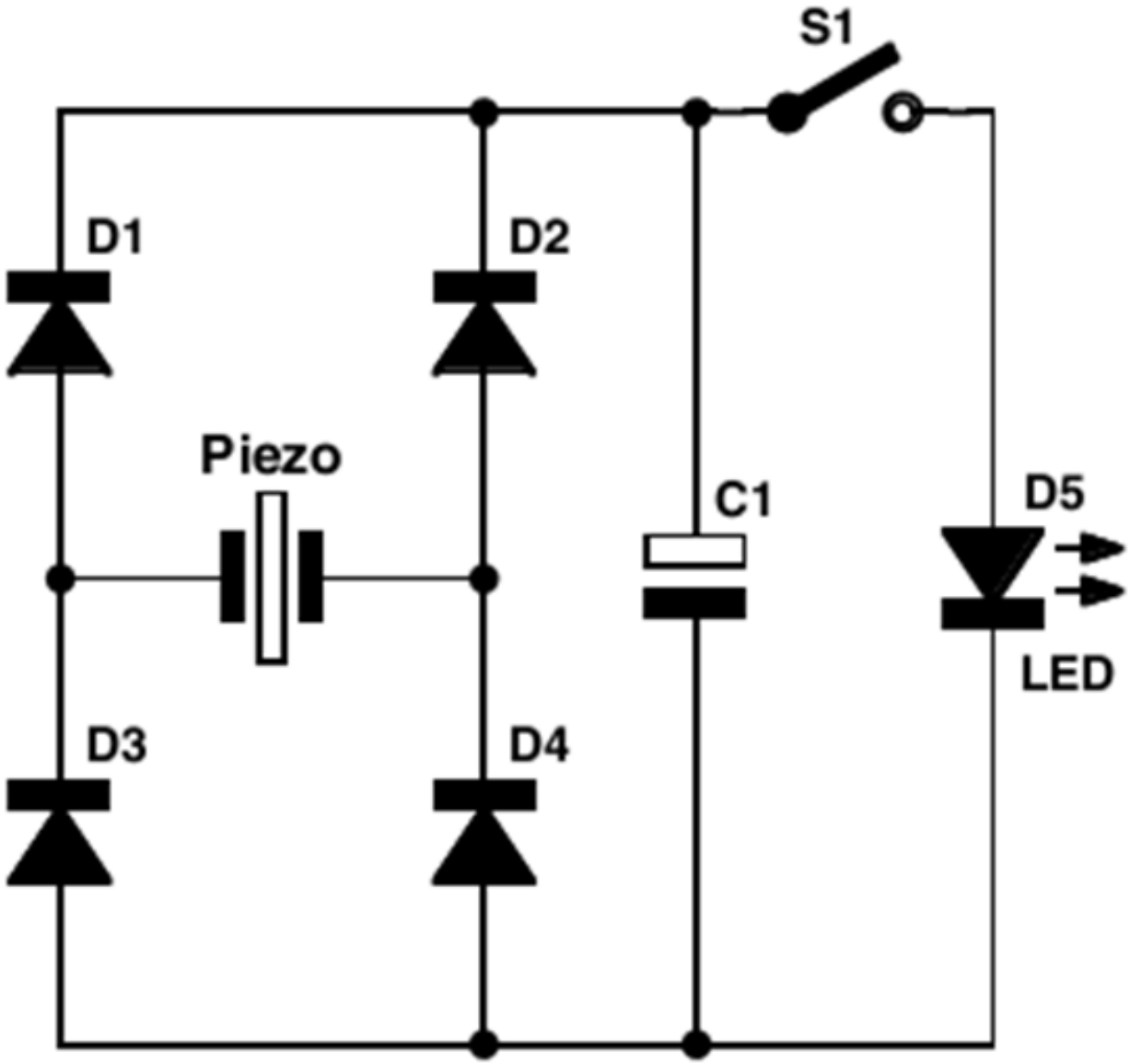




Circuito para Coletar Energia de Gerador Piezoelétrico

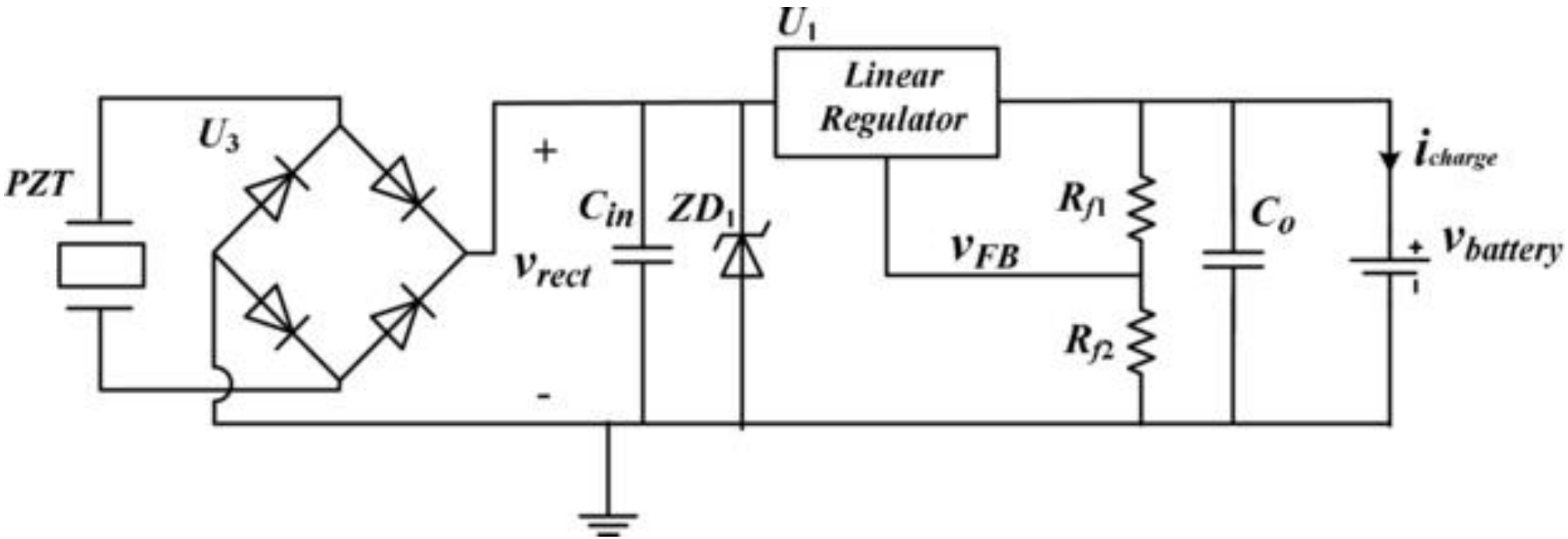


Ponte Retificadora para Coletar Energia de Gerador Piezoelétrico

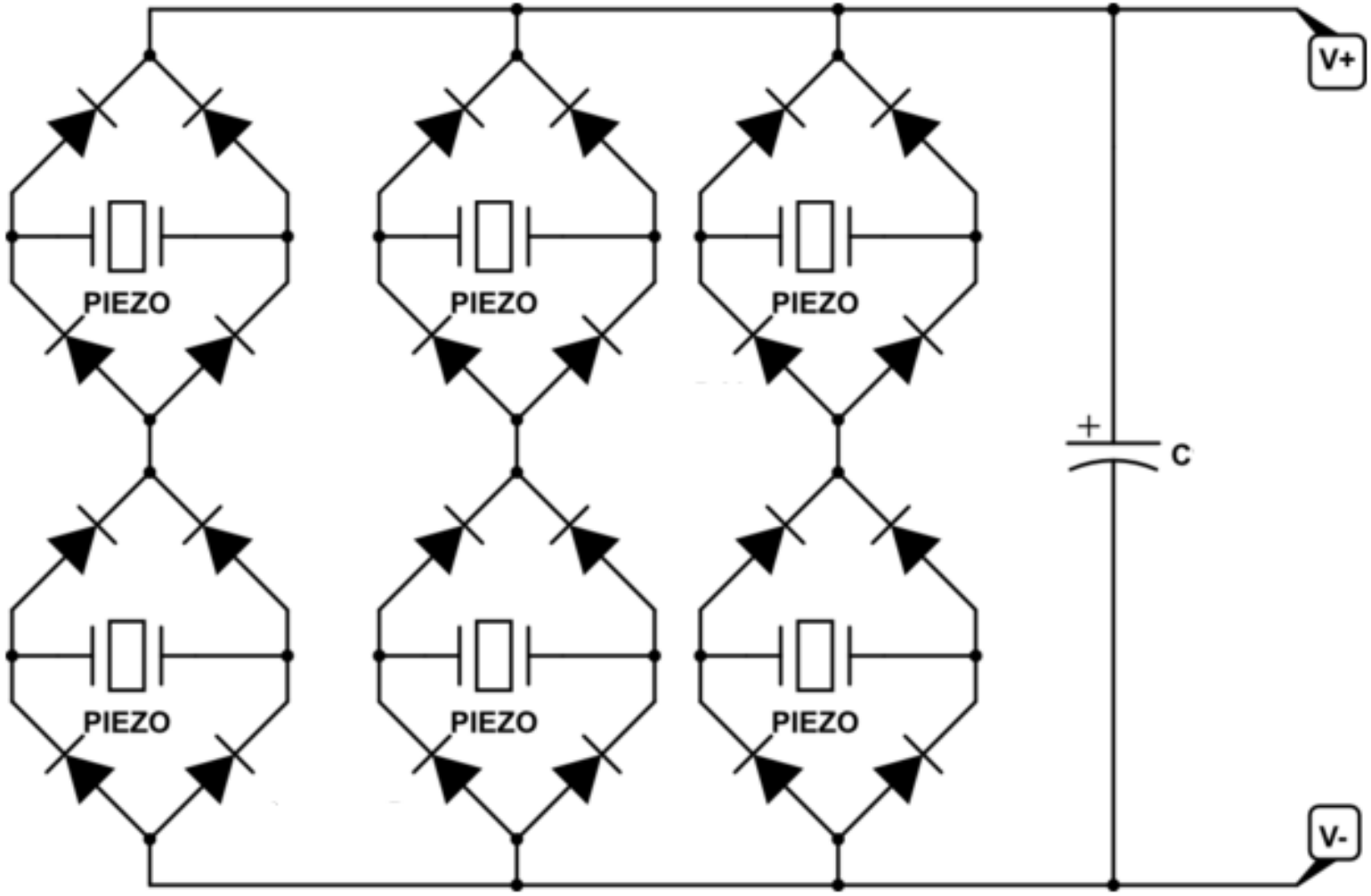


Author: Burkhard Kainka – Copyright: Revista Elektor

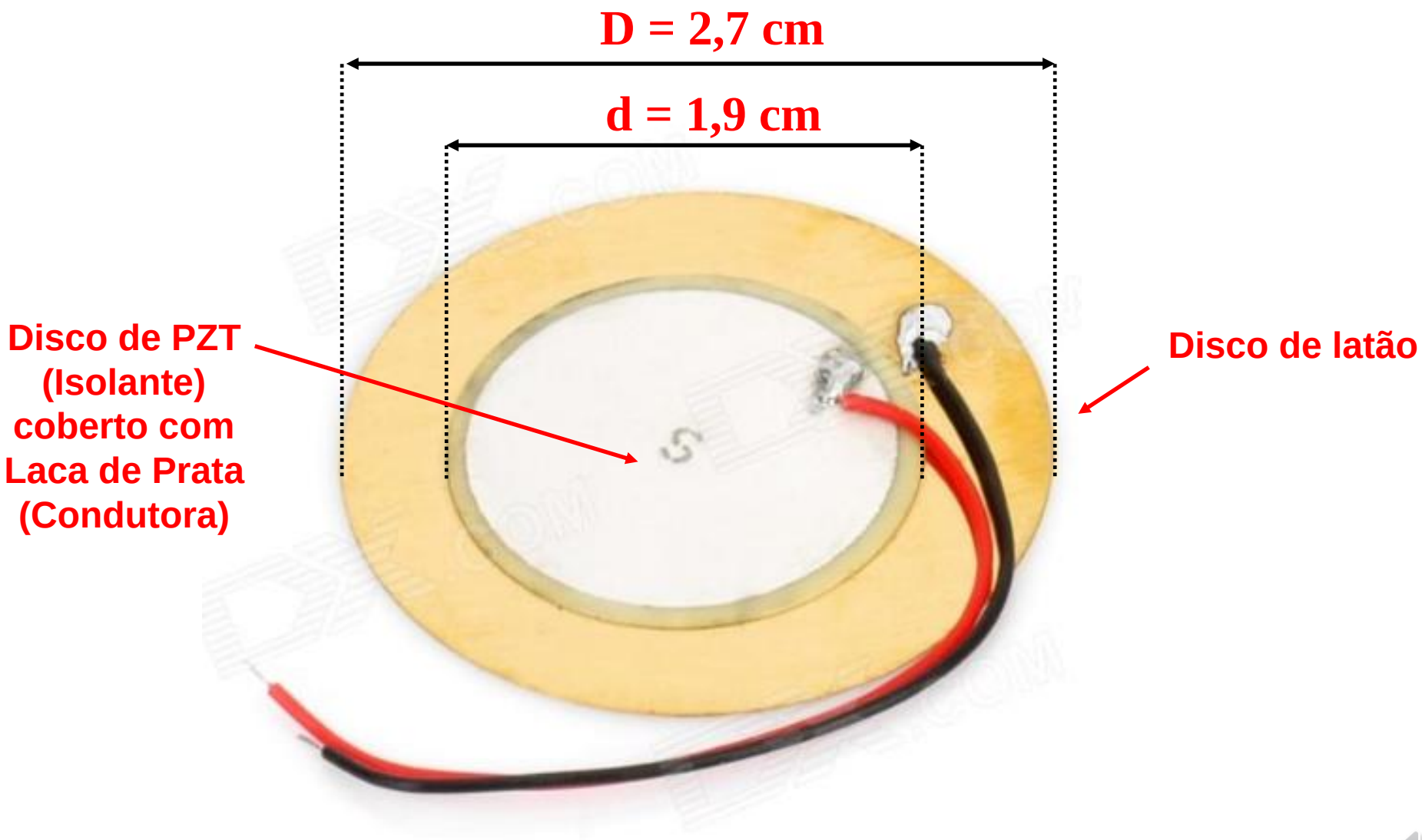
Circuito Regulado para Coletar Energia de Gerador Piezoelétrico



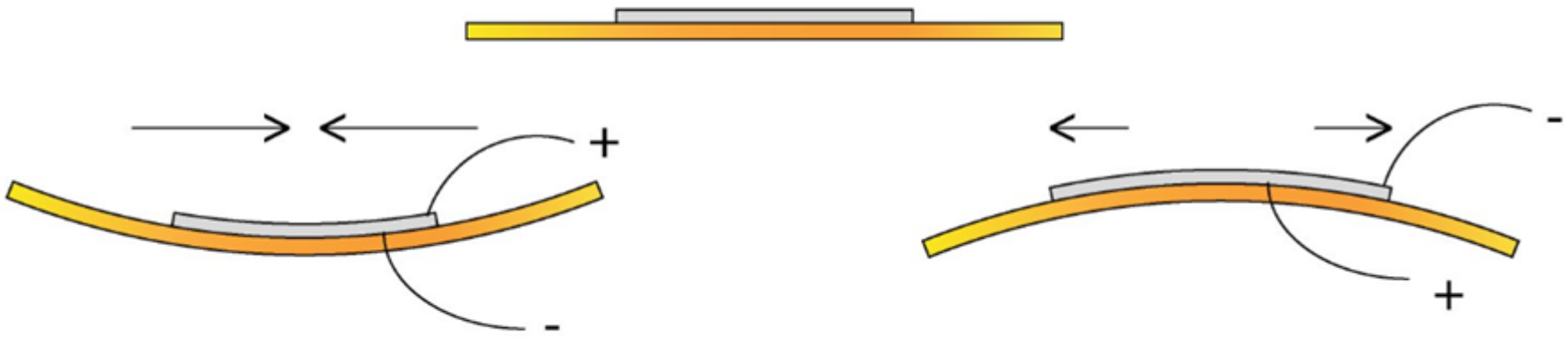
Array Gerador Piezoelétrico



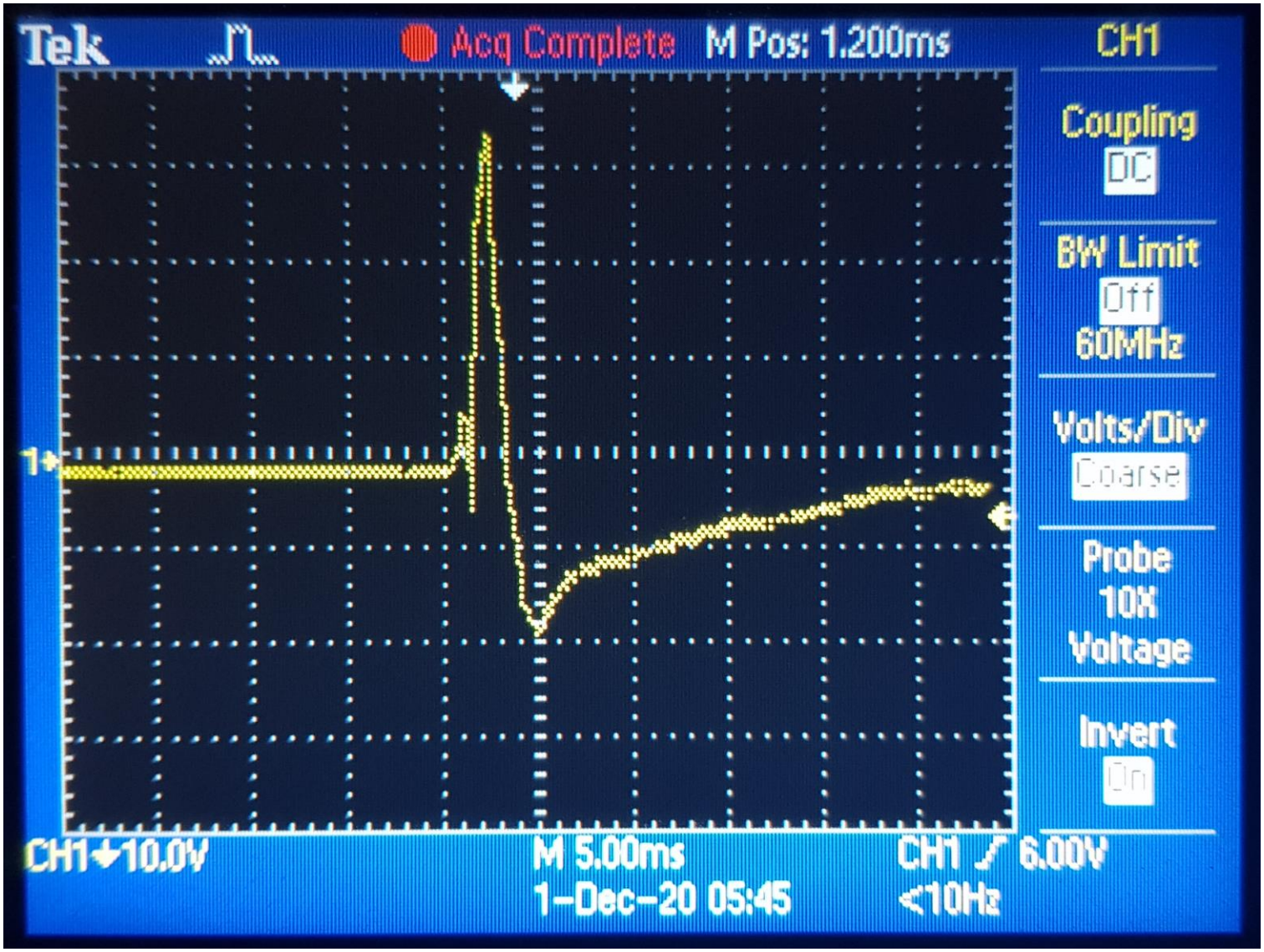
Disco Piezoelétrico



Disco Piezoelétrico



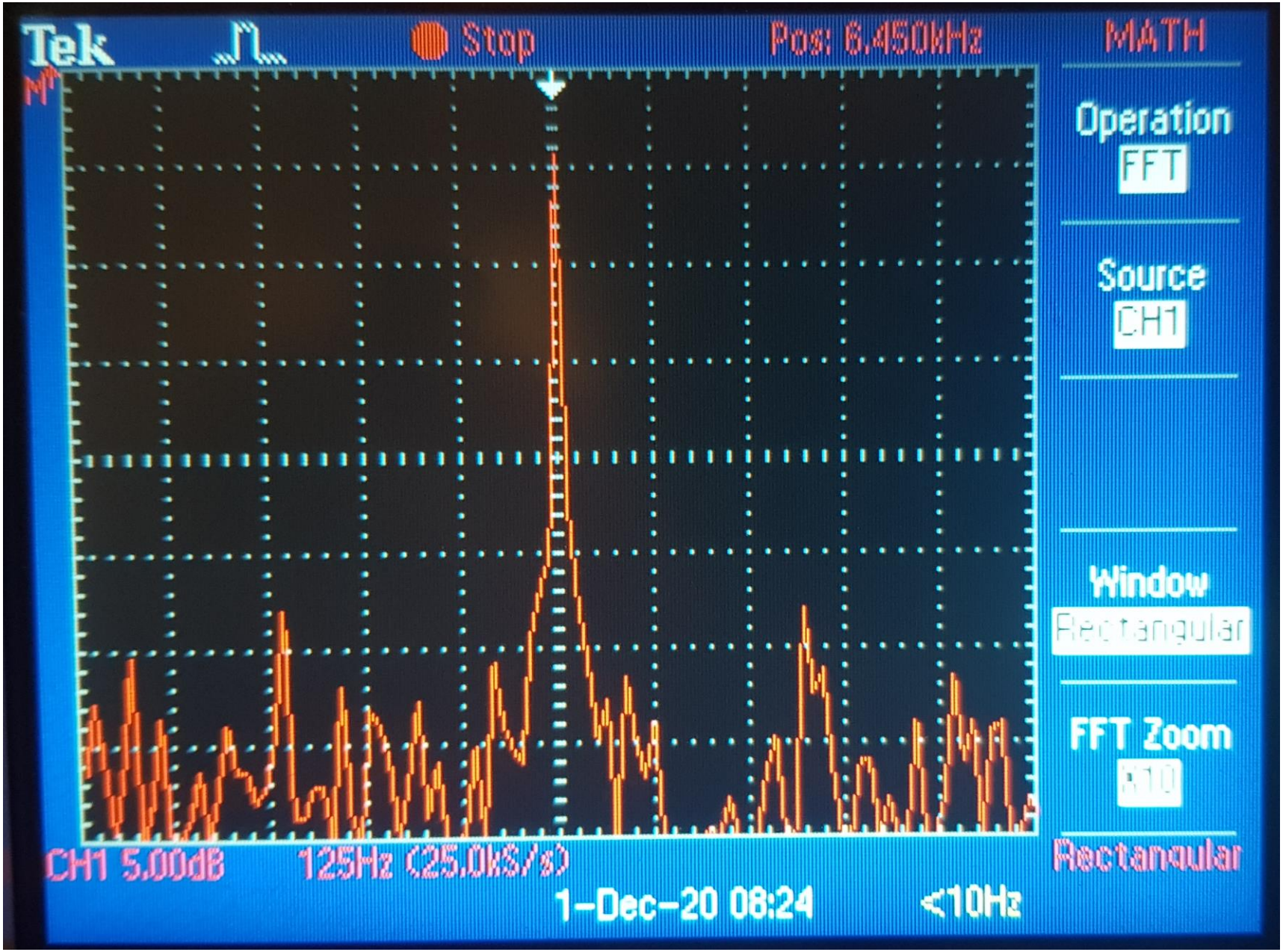
Procedimento Experimental: Parte-I (b)



V_P (Positivo) = **33** [volts]

V_P (Negativo) = **19** [volts]

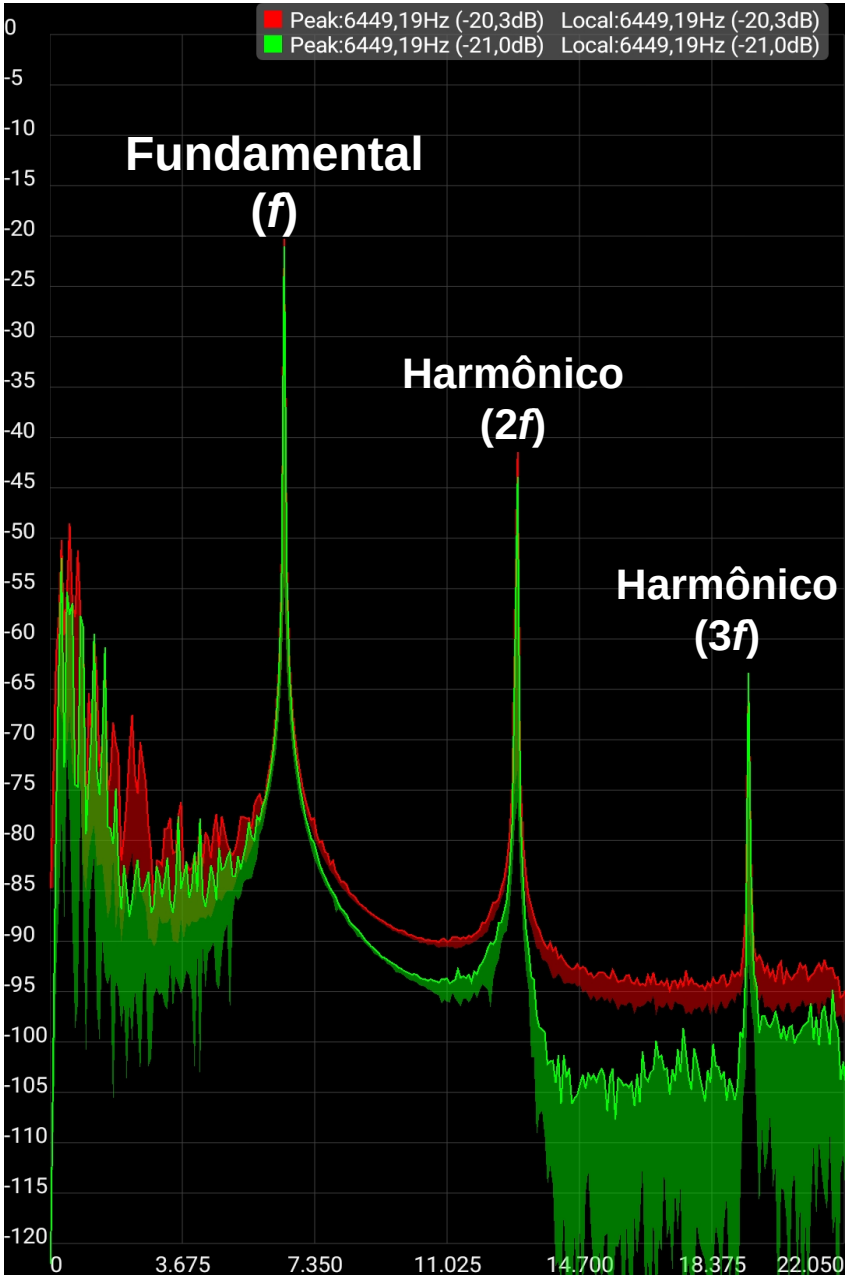
Experimento Experimental: Parte-I (c)



f = **6450** [Hz]

Procedimento Experimental: Parte-II

APP: Raspberrywood Spectrum Analyser



f (fundamental) = **6449** [Hz]

Procedimento Experimental: Parte-II

Ondas Sonoras: $v = \lambda f$

Discos Piezoelétricos:
(Eficiente Irradiador Sonoro)

$$D \approx \frac{1}{2}\lambda$$

Analogamente às Antenas do tipo Dipolo para ondas de Rádio.

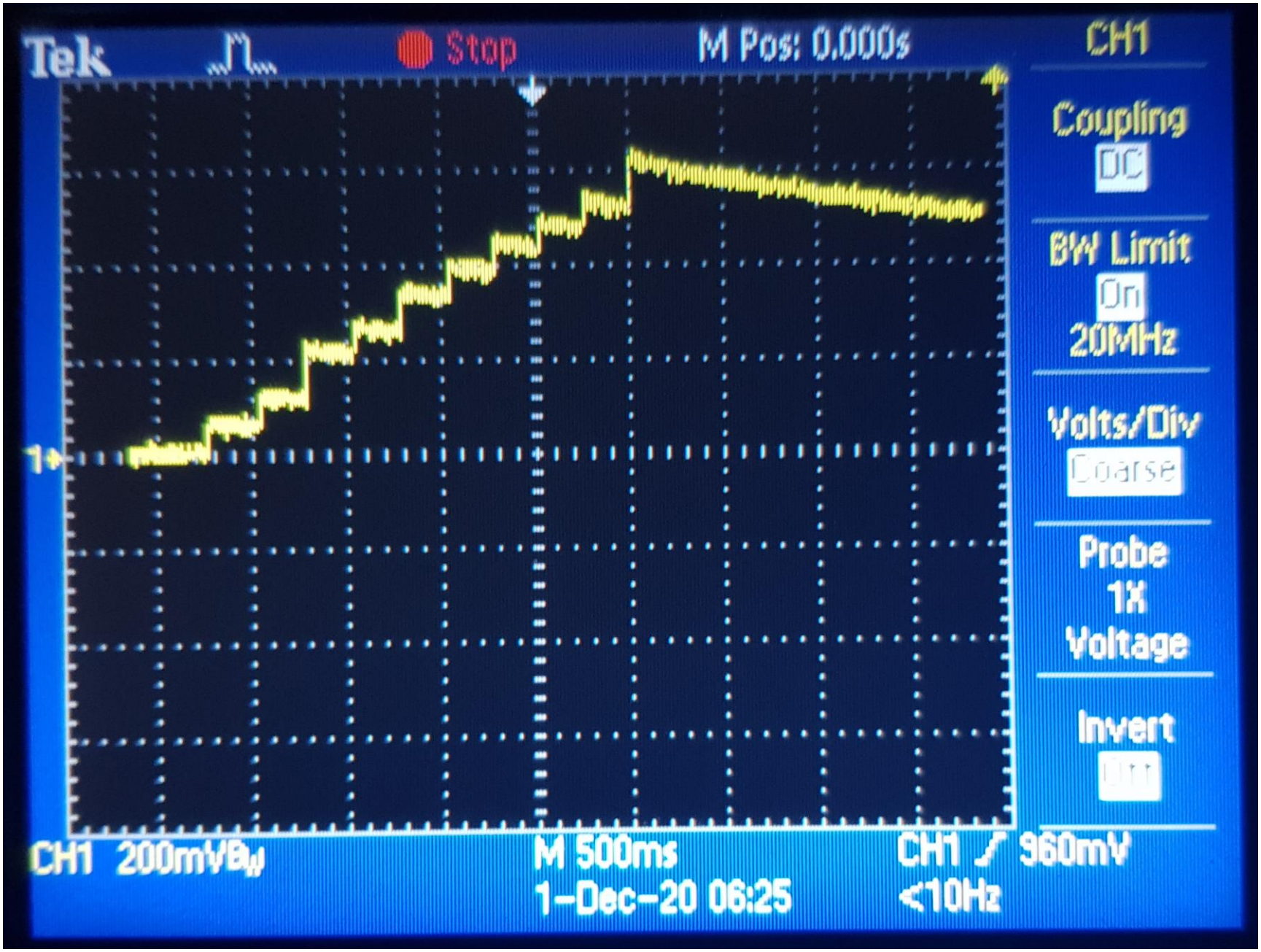
$$\lambda = \frac{v}{f} = \frac{340[m/s]}{6449[Hz]} = 0,0527[m]$$

$$\frac{D}{\lambda} = \frac{0,027[m]}{0,0527[m]} = 0,51 \approx \frac{1}{2}$$

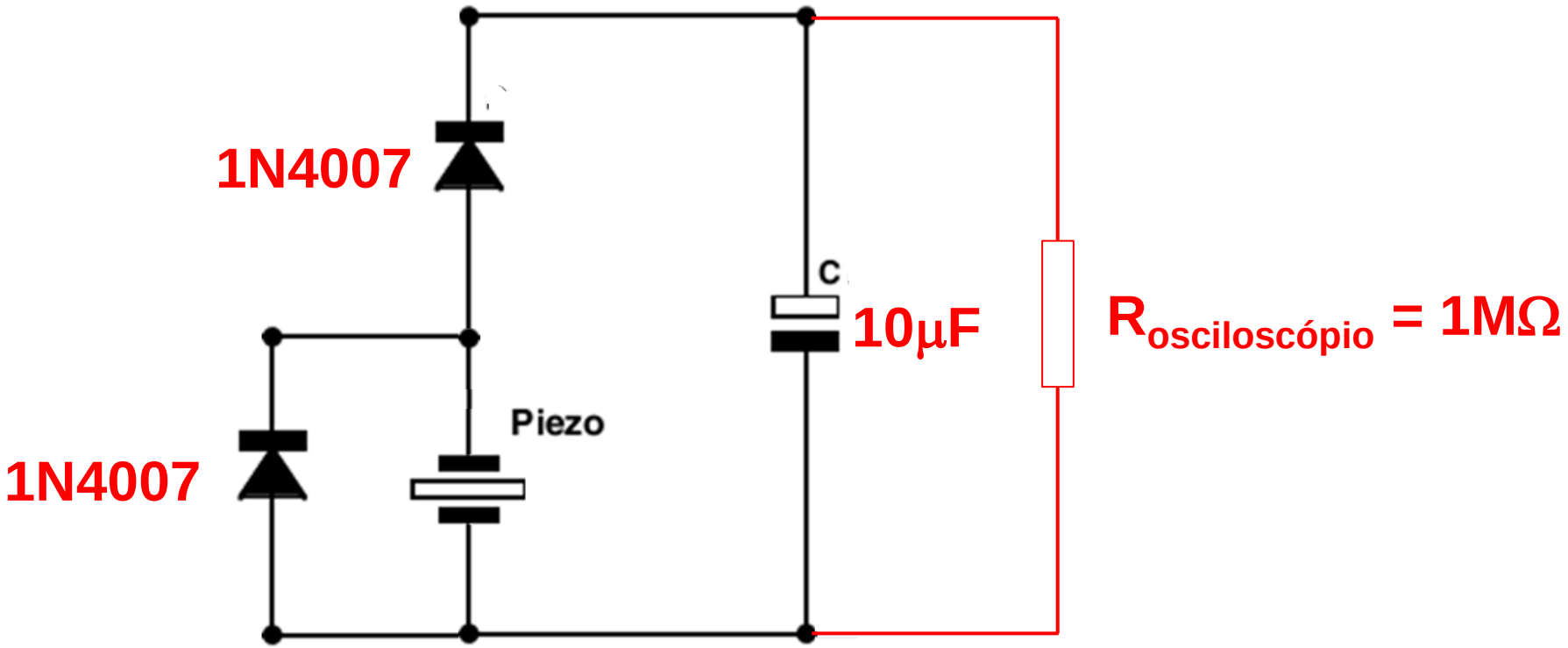
Observação: $f_{Parte-I(c)} \approx f_{Parte-II}$

O disco piezoelétrico é um eficiente irradiador e também é um bom sensor de ondas sonoras na mesma frequência!

Procedimento Experimental : Parte-III



Procedimento Experimental : Parte-III



$$E_{capacitor} = \frac{1}{2} C V^2$$

$$E_{capacitor} = \frac{1}{2}CV^2 = \frac{1}{2}(10\times10^{-6}F)(620\times10^{-3}volt)^2 = 1,9\times10^{-6}[J]$$

$$\boxed{E_{pulso} = \frac{E_{capacitor}}{10} = 1,9\times10^{-7}[J]}$$

Aproximação: $T_{total} << 2,2RC[s] = 22[s]$ **Tempo de Descarga (90% até 10%)**

$$T_{total} = 2500\times10^{-3}[s]$$

$$P_{média} = \frac{E_{capacitor}}{T_{total}} = \frac{1,9\times10^{-6}[J]}{2500\times10^{-3}[s]} = 7,6\times10^{-7}[W]$$